Inkloom AI

• FINAL HACKATHON PROJECT SUBMISSION

Smart Brand Checker & Publishing Platform

Inkloom AI helps companies review their marketing text and pitch decks before publishing. It catches banned words, checks facts, fixes tone in one click, and lets you safely publish to social media and websites.

TEAM DETAILS

SATYACODEZ

FutureStack Hackathon

TEAM LEADER

Satyam Kumar

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SOURCE CODE

GitHub Public Repository

satya-no17/futureStack

Everything Built and Working in the Product

A Complete System That Keeps Company Content Accurate, On-Brand, and Ready to Publish

01

Catch Mistakes Early

Reviews your draft before anything is published to the public.

- Catches banned words and risky promises
- Checks tone so you sound professional
- Verifies facts against official documents
- Ensures sentences are easy to read

Instant
FAST AUTOMATIC REVIEW

02

1-Click Smart Fixes

Highlights the exact wrong words and suggests approved replacements.

- Replaces only the problem words
- Keeps your own personal writing style
- Click once to apply the approved phrase
- Updates your compliance score immediately

1 Click
TO APPLY APPROVED FIX

03

Team Review Queue

A shared space where writers and managers collaborate cleanly.

- Authors submit drafts for review
- Managers approve or request changes
- Red and green text highlights show diffs
- Full audit log of who approved what

3 Roles
CREATOR, REVIEWER, ADMIN

04

All 3 Upgrades Shipped

All major improvements promised for Round 3 are fully delivered.

- Pitch Deck Scanner for multi-slide decks
- 1-Click Social Media Publishing
- Webhooks for WordPress & Webflow
- Self-Healing Brand Rules Engine

3 of 3
COMMITMENTS DELIVERED

✓ **Live on the Dashboard: Writers, reviewers, and admins can test all 8 modules directly in the app.**

WORKING IN LIVE PRODUCT

The 4-Step Review Process

Every Draft Goes Through 4 Clear Checks to Catch Different Types of Mistakes

Step 1 • Fast Filter

Banned Words

Catches prohibited terms and risky promises in under a second.

- Flags words like "cheap", "guaranteed", or "#1"
- Highlights the exact word position
- Labels risk: High, Medium, or Low
- Suggests the approved company word

Step 2 • Voice Check

Brand Voice & Tone

Checks if the writing sounds like your company's personality.

- Flags overly casual slang and hype
- Makes sure tone is confident and helpful
- Checks against company style guidelines
- Explains clearly why the tone feels off

Step 3 • Fact Checker

Truth & Claim Check

Verifies numbers and claims against official company documents.

- Flags unverified statistics or pricing
- Confirms promises like 99.95% uptime
- Labels: Supported, Unverified, or False
- Cites the official source document

Step 4 • Readability

Easy to Read & See

Makes sure anyone can easily read the text and see graphics.

- Scores reading grade level (easy to read)
- Warns about overly long run-on sentences
- Tests image contrast for readability
- Calculates an overall score from 0 to 100

✓ **Total Review Time: Fast local checks run in milliseconds, followed by smart AI analysis in just 1 to 2 seconds.**

FAST & THOROUGH

Writer Studio: See Mistakes & Fix Them in 1 Click

We Help Writers Fix Off-Brand Words Without Destroying Their Original Message

Color-Coded Visual Highlighter

Writers type naturally. When they click "Check My Words", the app highlights problem words in clear colors:

What the writer sees in the editor:

"Our **cheap cloud** is **guaranteed** to work with **instant** setup."

- **Red Highlight:** Banned Word ("cheap" sounds low quality)
- **Yellow Highlight:** Unverified Claim ("guaranteed" has legal risks)
- **Orange Highlight:** Wrong Tone ("instant" is misleading hype)
- Click any colored phrase to see why it was flagged and how to fix it.

Helpful Suggestion Cards with 1-Click Fix

Instead of rewriting your whole paragraph, Inkloom suggests a clean, approved replacement that fits right in:

MUST FIX • BANNED WORD

Rule: Brand Vocabulary Policy

Problem phrase: “cheap cloud storage”

The word 'cheap' makes our enterprise cloud sound low quality.

Suggested: “cost-effective cloud storage” 1-Click Apply

- Your brand score updates immediately when you accept a fix.
- If you disagree, you can dismiss it with an explanation note.
- The writer stays in complete control of their article.

Team Review Queue & Before-and-After Viewer

Managers Can Review, Compare Changes, and Sign Off in Seconds

Editorial Review Queue

When writers finish a draft, they send it to the review queue. Managers can see all pending work in one organized place:

- **Approve:** Signs off on good drafts and marks them ready to publish.
- **Request Changes:** Sends clear notes back to the writer with highlighted mistakes.
- **Line-by-Line Comments:** Leave helpful notes directly on specific paragraphs.
- **Full Audit History:** Keeps an automatic record of who approved what and when.

100%
LOGGED AUDIT HISTORY

Instant
HANDOFF TO REVIEWER

Before & After Changes Viewer

Reviewers can instantly see what changed between revisions with red and green text highlights. No need to re-read everything:

REMOVED IN REVISION (RED):

```
- "Our crazy fast platform is the cheapest on the market guaranteed."
```

APPROVED REVISION (GREEN):

```
+ "Our platform delivers sub-50ms latency backed by our 99.95% uptime SLA."
```

- Quickly check if the writer followed previous review feedback.
- Save hours of proofreading by focusing only on changed text.

ROUND 3 COMMITMENT 1 DELIVERED

Pitch Deck & Slide Scanner

Delivered: Upload Presentation Decks to Check Slide Margins, Typography, and Claims

1. Reads Multi-Slide Decks

Upload whole pitch decks. Inkloom reads each slide individually and displays visual slide thumbnails:

- Supports structured decks and presentation files
- Calculates a safety score for each slide
- Click any slide thumbnail to inspect it
- Quickly see which slides have issues

2. Logo Margins & Layout

Checks visual space around logos and warns if text is messy or violating branding:

- **Logo Clearance:** Flags logos placed too close to edges (needs 20px space)
- **All-Caps Check:** Flags aggressive SHOUTING headlines
- **Crowding:** Flags overlapping text and images

3. Checks Claims on Slides

Compares bullet points on slides against verified company technical spec sheets:

- Flags exaggerated claims like "#1 fastest"
- Verifies uptime claims against SLA sheets
- Gives slide-by-slide fix suggestions
- Ensures investor decks are factually safe

✓ **Tested in Automated Test Suite: Tested a 3-slide pitch deck and caught 13 margin, font, and claim mistakes.**

VERIFIED WORKING

ROUND 3 COMMITMENT 2 DELIVERED

1-Click Social Media & Webhook Publishing

Delivered: Publish Approved Content Straight to Social Media Without Extra Setup

Zero-Setup Social Media Sharing

Judges and users can test publishing immediately. No developer accounts, OAuth setup, or secret keys needed:

- **Twitter / X:** Formats the approved draft, checks the 280-character limit, and opens a 1-click posting window.
- **LinkedIn:** Opens a pre-filled share window ready to post to your profile or company page.
- **Proof Watermark:** Adds an optional badge: `[Verified by Inkloom AI — Score 100/100]`.

7 Channels
SOCIAL & CMS TARGETS

Zero Setup
WORKS OUT OF THE BOX

Webhooks for Websites & Blogs

Automatically sends approved articles and compliance reports directly to websites (WordPress, Webflow) or workflows (Zapier, Slack):

```
// Webhook data sent when you hit Publish:
{
  "event": "content.published",
  "title": "Enterprise Cloud Announcement",
  "score": 100,
  "status": "approved",
  "publishedBy": "reviewer@company.com"
}
```

- Sends text, score, and author info in clean data.
- Has a 10-second safety timeout and delivery verification.

Self-Healing Brand Rules Engine

Delivered: When Writers Keep Breaking a Rule, AI Automatically Rewrites It in Plain English

1. Finds Confusing Rules

Looks at history logs to find rules that multiple team members keep failing over and over again:

- Counts violation frequency per rule
- Spots when multiple writers struggle
- Detects when suggestions are dismissed
- Surfaces the problem rule on the Dashboard

2. AI Suggests Clearer Rule

Uses AI to rewrite the vague rule in simple English, adding clear "Do" and "Don't" examples:

- Replaces confusing jargon with clear instructions
- Provides concrete good and bad examples
- Explains why writers were confused
- Presents a ready-to-approve suggestion

3. 1-Click Rulebook Update

Brand managers review the suggestion and click "Apply Clarification" to update the company rulebook:

- Instantly updates active company rules
- Keeps full history of all rule changes
- All future scans immediately enforce the new rule
- No manual rule writing needed

✓ **Live on the Score & Reports Tab: SelfHealingCard displays active suggestions with a 1-click button to update the rules.**

VERIFIED WORKING

How the System is Built

Modern, Reliable Web Technologies Connected to Fast AI and Vector Search

Frontend (User Interface)

Next.js + React + Tailwind CSS

A clean, modern interface that loads fast, handles draft editing, slide uploading, and works smoothly on desktop and mobile.

- Fast page loading with Next.js App Router
- Clean card layout and icons
- Inline text highlighter engine
- Step-by-step onboarding walkthrough

Backend (API & Rules)

Node.js + Express API

Runs instant rule checks in milliseconds, parses slide decks, coordinates AI reviews, and delivers fast results.

- Fast REST API endpoints
- Sub-50ms instant regex checks
- Slide deck and XML layout reader
- Secure password and role checks

Database & Search

PostgreSQL with Vector Search

Stores team drafts, review comments, and uses fast vector similarity search to find matching brand rules in milliseconds.

- Fast similarity search for brand rules
- Safe draft and revision history
- Multi-role team permissions
- Full audit log tracking

✓ Resilient Failover: Uses Google Gemini 1.5 Flash for tone reviews. If internet or AI is unavailable, the app uses built-in local rules and demo records so writers are never blocked.

ZERO WHITE-SCREEN ERRORS

Testing Results & Future Roadmap

100% Passed Across 42 Automated Scenarios and Clear Next Steps

Testing: 100% Pass Rate (42/42 Tests)

Tested thoroughly with an automated test suite covering 4 key areas:

- **Feature Checks (24/24 PASS):** Every button, walkthrough step, demo card, and color contrast passed.
- **Stress & Bad Inputs (8/8 PASS):** Handled network drops, invalid passwords, and giant 50,000-character inputs smoothly.
- **Offline Reliability (6/6 PASS):** Every single tab in the app works even with internet or database disconnected.
- **Complete User Journeys (4/4 PASS):** Tested complete writer, reviewer, and admin flows from start to finish.

42 / 42 PASS (100%)
AUTOMATED VERIFICATION SCENARIOS PASSING

Where It Goes Next (Product Roadmap)

What we will build next and who it is for:

- **1. Browser Extension for Google Docs & Notion:** Content creators can see brand warnings right inside Google Docs, Notion, or Canva while typing.
- **2. Two-Way Website Sync (WordPress & Webflow):** For marketing teams—automatically updates live blog posts when brand rules change.
- **3. Multi-Language Brand Voice:** For global companies—ensures consistent tone across English, Spanish, French, and Japanese.
- **4. Enterprise Single Sign-On (SSO):** For large companies needing Okta, Google Workspace, and Microsoft Azure login.

4 Next Steps
CLEAR PRODUCT ROADMAP